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Description

This week, I learned about decision trees as a supervised machine learning method for classification and regression. In the practical activity, I used a Wine Classification dataset and learned how a decision tree splits data using features and thresholds. I also learned about nodes, branches, leaves, tree depth, Gini impurity, and entropy.

Feelings

At first, I found decision trees relatively easy to understand because their structure is similar to asking a series of questions. However, I felt less confident when learning how Gini impurity is calculated and how the algorithm selects the best split. This made me realise that understanding the mathematical reasoning behind a model is more challenging for me than understanding its overall structure.

3. Evaluation

A positive aspect of the practical activity was that I could connect the theory of decision trees with Python implementation. Using `DecisionTreeClassifier`, splitting the data into training and testing sets, and visualising the tree made the modelling process clearer and more practical. A negative aspect was that I sometimes focused on getting the code to run rather than understanding why the model made particular decisions. I also found calculating Gini impurity more difficult than the programming itself.

Analysis

This activity improved my technical skills in machine learning and Python, but it also highlighted an area for improvement in my problem-solving skills. A model can produce predictions successfully without me fully understanding how those predictions are generated. I therefore need to develop the habit of interpreting model structure and performance rather than relying mainly on code output. This will help me make more informed decisions when selecting and evaluating models.

Conclusion

Overall, I developed a stronger understanding of how decision trees make predictions and how their performance can be affected by overfitting. I also became more aware that becoming a data scientist requires both programming ability and the ability to interpret and critically evaluate model results.

6. Action Plan

In future, I will use the Wine dataset to build decision trees with different maximum depths and compare their training and testing accuracy. I will also calculate the Gini impurity for one split by hand and check it against Python's result. This will help me better understand the algorithm and identify overfitting.

